

Chapter 5.1: Introduction

Memory Hierarchy

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1 Introduction: The Dream of Unlimited Fast Memory

“From the earliest days of computing, programmers have wanted unlimited amounts of fast memory.”

This chapter explores how computer systems create the **illusion** of having both:

- **Unlimited memory capacity** (like a massive library)
- **Very fast access speed** (like having a book already in your hand)

2 The Library Analogy

Imagine you are a student writing a research paper in a library:

- You have a **desk** with a few books you’ve pulled from the shelves.
- Most of the time, you find what you need in the books **already on your desk**.
- Occasionally, you need to go back to the **shelves** to retrieve a new book.
- Having several books on your desk **saves time** compared to constantly walking back to the shelves.

Library ↔ Computer Mapping

Library Concept	Computer Concept
Books on your desk	Cache (fast, small)
Library shelves	Main Memory (slower, larger)
Finding info in a book on your desk	Hit
Going to shelves for a new book	Miss
A single book	A Block (or Line)

3 The Principle of Locality

The Core Principle

The Principle of Locality states that programs access a **relatively small portion** of their address space at any instant of time.

This principle is the foundation that makes memory hierarchies effective. There are two types:

3.1 1. Temporal Locality (Locality in Time)

Temporal Locality

If a data item is referenced, it will tend to be **referenced again soon**.

Example: If you recently brought a book to your desk, you will probably need to look at it again soon.
In Programs:

- **Loops:** Instructions within a loop are executed repeatedly.
- **Variables:** A variable used in a calculation is often used again soon after.

3.2 2. Spatial Locality (Locality in Space)

Spatial Locality

If a data item is referenced, items whose **addresses are close by** will tend to be referenced soon.

Example: When you got a book on early British computers, you noticed a related book next to it on the shelf and brought that too.

In Programs:

- **Sequential instruction execution:** After fetching instruction i , we very likely fetch instruction $i + 1$.
- **Array access:** Accessing `array[0]` often means `array[1]`, `array[2]`, etc., will be accessed soon.

4 The Memory Hierarchy

Memory Hierarchy

A structure that uses **multiple levels of memory** with different speeds and sizes. As the distance from the processor increases, the size of the memories and the access time **both increase**.

The key trade-off:

- **Faster memories** are more **expensive per bit** and thus smaller.
- **Larger memories** are slower but cheaper.

4.1 The Goal

Provide the user with as much memory as is available in the **cheapest technology**, while providing access at the speed offered by the **fastest memory**.

4.2 Structure of the Hierarchy

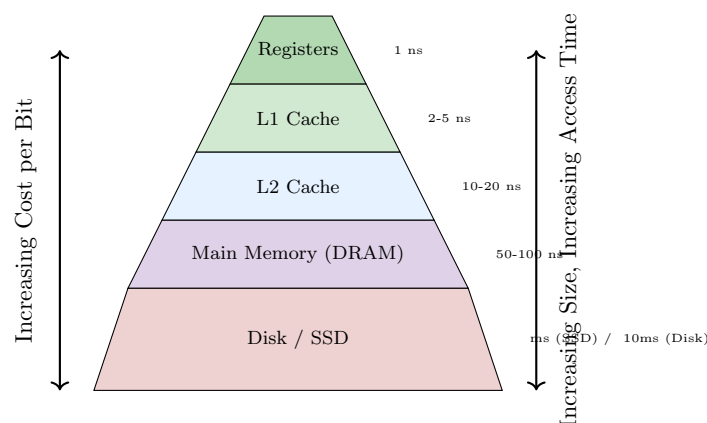


Figure 1: **Figure 5.1/5.3:** The basic structure of a memory hierarchy. Faster, smaller, more expensive memory is closer to the processor.

4.3 Key Properties

- Data in a level closer to the processor is generally a **subset** of data in levels further away.
- Data is copied between only **two adjacent levels** at a time.
- In a “true” hierarchy, data cannot be present in level i unless it is also present in level $i + 1$.

5 Key Terminology

5.1 Block (or Line)

Block (or Line)

The **minimum unit of information** that can be either present or not present in a cache.

When data is transferred between levels, we usually transfer an **entire block**, not just a single byte or word. This exploits spatial locality.

5.2 Hit and Miss

Hit

If the data requested by the processor **appears** in some block in the upper level, this is called a **hit**.

Miss

If the data **is not found** in the upper level, the request is called a **miss**. The lower level is then accessed to retrieve the block.

5.3 Hit Rate and Miss Rate

Hit Rate

The **fraction of memory accesses** found in the upper level of the memory hierarchy.

Miss Rate

The **fraction of memory accesses not found** in the upper level.

$$\text{Miss Rate} = 1 - \text{Hit Rate}$$

5.4 Hit Time and Miss Penalty

Hit Time

The time to access the upper level of the memory hierarchy, **including the time to determine** whether the access is a hit or a miss.

Miss Penalty

The time to **replace a block** in the upper level with the corresponding block from the lower level, **plus** the time to deliver this block to the processor.

Why is Miss Penalty so much larger than Hit Time?

The upper level is **smaller** and built using **faster memory parts**. The time to access the next lower level is the major component of the miss penalty.

Analogy: Looking at a book on your desk (hit time) is much faster than walking to the shelves, finding a book, and bringing it back (miss penalty).

6 The Big Picture

The Big Picture

- Programs exhibit **temporal locality** (reuse recently accessed data) and **spatial locality** (reference data near recently accessed items).
- Memory hierarchies exploit **temporal locality** by keeping recently accessed data closer to the processor.
- Memory hierarchies exploit **spatial locality** by moving **blocks** of multiple contiguous words to upper levels.
- If the **hit rate is high enough**, the memory system has:
 - An **effective access time** close to the fastest (highest) level.
 - A **size** equal to the largest (lowest) level.

7 Check Yourself

Which of the following statements are generally true?

1. **Memory hierarchies take advantage of temporal locality.**
TRUE. By keeping recently accessed data in fast, upper levels.
2. **On a read, the value returned depends on which blocks are in the cache.**
TRUE. If the data is in the cache (hit), it's returned from there. If not (miss), it comes from a lower level.
3. **Most of the cost of the memory hierarchy is at the highest level.**
*FALSE. The highest level (e.g., L1 cache) is the smallest, so its total cost is lower. Most cost is in the larger, lower levels. (Note: cost **per bit** is highest at the top, but **total cost** is dominated by lower, larger levels.)*
4. **Most of the capacity of the memory hierarchy is at the lowest level.**
TRUE. Disks and SSDs provide terabytes, while caches provide kilobytes or megabytes.

8 Flashcards

1. **Q: What illusion do memory hierarchies create for programmers?**
A: They create the illusion of having unlimited amounts of fast memory.
2. **Q: What fundamental principle underlies the operation of memory hierarchies?**
A: The principle of locality.
3. **Q: What does the principle of locality state about program behavior?**
A: It states that programs access a relatively small portion of their address space at any instant of time.
4. **Q: What are the two different types of locality?**
A: Temporal locality (locality in time) and spatial locality (locality in space).
5. **Q: Define Temporal Locality.**
A: The principle stating that if a data location is referenced, it will tend to be referenced again soon.
6. **Q: What common program structure demonstrates high temporal locality?**
A: Loops, where instructions and data are accessed repeatedly.
7. **Q: Define Spatial Locality.**
A: The principle stating that if a data location is referenced, data locations with nearby addresses will tend to be referenced soon.
8. **Q: How does sequential access of instructions in a program demonstrate spatial locality?**
A: Instructions are normally accessed sequentially, meaning that referencing one instruction makes it highly likely the next one in memory will be referenced soon.
9. **Q: Sequential access to elements of an array is an example of high _____ locality.**
A: Spatial.
10. **Q: How do computer systems take advantage of the principle of locality?**
A: By implementing the memory of a computer as a memory hierarchy.
11. **Q: Define Memory Hierarchy.**
A: A structure that uses multiple levels of memories where size and access time increase with distance from the processor.
12. **Q: What is the relationship between speed, size, and cost per bit in a memory hierarchy?**
A: Faster memories are more expensive per bit and are therefore smaller.
13. **Q: What is the overall goal of implementing a memory hierarchy?**
A: To provide memory as large as the cheapest level, with an access speed offered by the fastest level.
14. **Q: In a memory hierarchy, a level closer to the processor is generally a _____ of any level further away.**
A: Subset.
15. **Q: In a memory hierarchy, data is copied between only _____ adjacent levels at a time.**
A: Two.
16. **Q: Define Block (or Line).**
A: The minimum unit of information that can be either present or not present in a level of the memory hierarchy.
17. **Q: What is it called when data requested by the processor is found in the upper level of the memory hierarchy?**
A: A hit.
18. **Q: What is it called when data requested by the processor is not found in the upper level of the memory hierarchy?**
A: A miss.
19. **Q: What action is taken in a memory hierarchy when a miss occurs?**
A: The lower level in the hierarchy is accessed to retrieve the block containing the requested data.
20. **Q: Define Hit Rate.**
A: The fraction of memory accesses found in a level of the memory hierarchy.

21. **Q: Define Miss Rate.**
A: The fraction of memory accesses not found in a level of the memory hierarchy.
22. **Q: How is the miss rate related to the hit rate?**
A: The miss rate is equal to 1 minus the hit rate.
23. **Q: Define Hit Time.**
A: The time required to access a level of the memory hierarchy, including the time to determine if the access is a hit or a miss.
24. **Q: Define Miss Penalty.**
A: The time required to replace a block in an upper level with the corresponding block from the lower level and deliver it.
25. **Q: Why is the hit time much smaller than the miss penalty?**
A: Because the upper level is smaller and built with faster memory parts than the lower level.
26. **Q: Memory hierarchies keep more recently accessed data closer to the processor to take advantage of _____ locality.**
A: Temporal.
27. **Q: How do memory hierarchies leverage spatial locality?**
A: By moving blocks consisting of multiple contiguous words in memory to upper levels of the hierarchy.
28. **Q: In a true memory hierarchy, if data is present in level 'i', where else must it also be present?**
A: It must also be present in the next lower level, 'i + 1'.
29. **Q: With a high enough hit rate, a memory hierarchy has an effective access time close to that of the _____ level.**
A: Highest (and fastest).
30. **Q: A memory hierarchy gives the user the illusion of a memory size equal to that of the _____ level.**
A: Lowest (and largest).
31. **Q: In the library analogy for memory hierarchies, what do the books on your desk represent?**
A: The upper, faster level of the memory hierarchy (e.g., a cache).
32. **Q: In the library analogy, going back to the shelves to find a book corresponds to what event in a memory hierarchy?**
A: A miss, which requires accessing the lower level of memory.
33. **Q: In the library analogy, what does finding the information you need in a book already on your desk represent?**
A: A hit in the memory hierarchy.
34. **Q: In the library analogy, what does a single book represent?**
A: A 'block' or 'line' of information.
35. **Q: What is the major component of the miss penalty?**
A: The time to access the next, lower level in the hierarchy.
36. **Q: Why must modern programmers understand the memory hierarchy?**
A: To write code that achieves good performance, as memory access is a major factor in determining performance.